Production Planning and Scheduling Project

Aurelius SA



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Today's Agenda

01 Aggregate Planning

02 Inventory Management

03 Lot-Sizing Models

04 MRP

05 Base scheduling

06 Disruptions

07 Comparisons of disruptions

PART 1

Aggregate Planning

*In the project we determined planning strategies according to three different strategies : **Level, Chase and MILP**. They resulted in the following total costs :*

Strategy	Total Cost
Level	32,700,000
Chase	29,805,000
MILP 1/3	20,280,000
MILP 2/4	19,607,000

In order to fulfill demand, while minimizing its costs, Aurelius SA should apply the following production strategy :

```
--- Monthly Plan ---
Month  Demand  Workforce  Hired  Fired  Production  Overtime  Ending Inventory
  Jan     900         80      0     10       800.0       0.0               800
  Feb     950         80      0      0       800.0       0.0               650
  Mar    1200         80      0      0       800.0       0.0               250
  Apr    1050         80      0      0       800.0       0.0                 0
  May    1100        115     35      0      1150.0       0.0                50
  Jun    1300        125     10      0      1250.0       0.0                 0
  Jul    1250        125      0      0      1250.0       0.0                 0
  Aug    1100        125      0      0      1250.0       0.0               150
  Sep    1300        126      1      0      1260.0       4.0               114
  Oct    1450        173     47      0      1730.0       0.0               394
  Nov    1500        173      0      0      1730.0       0.0               624
  Dec    1700        173      0      0      1730.0     346.0              1000
```

Subcontracting (15,000 CHF/watch): is never used as it is far too expensive compared to regular production (700 CHF/watch).
Overtime: is only used marginally in September and December when hiring permanent workers would cost more.
The optimizer therefore adjusts workforce and depletes inventory to zero mid-year to minimize holding costs, then rebuilds it in the final quarter to meet the 1,000 watch target.
While cost-optimal, this leaves no buffer during mid-year, a risk for a luxury watchmaker facing unexpected demand spikes.

```
--- Annual Cost Breakdown ---
  Wage Cost     :   10,185,000 CHF
  Hire Cost     :    4,650,000 CHF
  Fire Cost     :      250,000 CHF
  Holding Cost  :    4,032,000 CHF
  Overtime Cost :      490,000 CHF
  ------------------------------
  TOTAL         :   19,607,000 CHF
```

PART 1

Inventory Management

- *Most components show stable demand → EOQ-based policy is appropriate*
- *(Q,R) Continuous Review recommended for high-value components (Diamond Marker, 3235 Movement) most critical and costly - (S,T)*
- *Periodic Review sufficient for low-value components (Jubilee Bracelet, Lubricant Oil, Sapphire Crystal)*

```
=== Annual Inventory Costs per Component ===

                Component  EOQ  Orders/Year  Ordering Cost/Year  Holding Cost/Year  Total Cost/Year (CHF)
   3235 Movement (units)   81         64.5                580762            1234133                1814895
Oystersteel Case (units)  127         43.6                305077             531820                 836897
Sapphire Crystal (units)  131         39.4                157790             235359                 393149
Jubilee Bracelet (units)   88         33.2                116267             141812                 258080
   Lubricant Oil (grams)  428        128.5                128469             280534                 409002
  Diamond Marker (units)  139        125.3                751896            1166578                1918474
      Gold Crown (units)   83         35.0                175160             307670                 482830
```

- ***Key lever:** reducing lead times (especially 3235 Movement: 60 days) would significantly reduce safety stock and total inventory costs*

PART 2

Lot-Sizing Models

The optimal Lot-Sizing Model for Aurelius SA, was first determined on a theoretical basis with the LS-U model, and then the realistic model was established with the LS-C model.

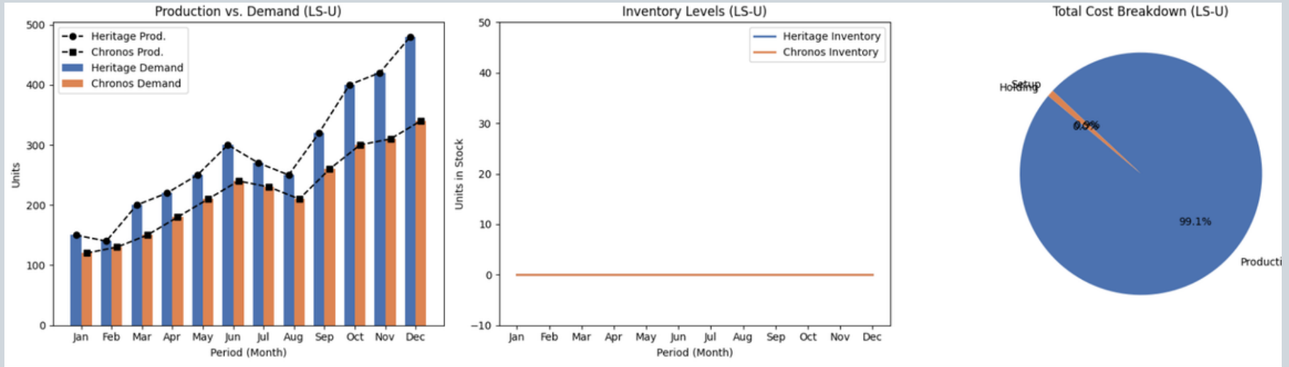
	LS-U	LS-C	Difference
Total Cost	**59,900,000**	**60,118,000**	**218,000**
Produced in Periods	**Both : 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12**	**Both : 1, 2, 3, 4, 5, 7, 8, 9, 11**

LS-U:

- Strict Lot-for-Lot strategy :
  - production exactly matches demand
  - zero-inventory and monthly setups
- Cause : holding costs are highly punitive compared to setup costs.

LS-C:

- Adding constraints => L4L approach no longer viable
- strategic shift to anticipation inventory :
  - Zero-production during peak months
  - Pre-building of inventory to meet demand
    - Cheaper than producing during peak-months



PART 2

MRP

For this analysis, we proceeded in two steps: 1. Sequential MRP and 2. Integrated MRP.

The outputs of the MRP drive three major business benefits :

- **Proactive Risk Mitigation:** Rather than reacting to shortages, managers receive a precise procurement schedule.
- **Strategic Sales Arbitrage:** The model gives managers a clear financial justification to accept or reject external client orders, ensuring that B2B sales never jeopardize the company's internal watch assembly.
- **Financial Visibility:** By dynamically balancing massive setup fees against monthly holding costs, the MRP establishes the most cost-efficient baseline for the entire year, directly protecting and maximizing Aurelius SA's profit margins.

Total costs	No external demand	External demand	Difference
Sequential	129,575,010	129,589,570	14,560
Integrated	129,424,860	129,692,970	268,110
Number of setups	sequential : 5 integrated : 5	sequential : 6 integrated : 5	sequential : 1 integrated : 0
Difference	150,150	103,400	

Highly efficient, yet fragile operation system : Lack of operational elasticity to absorb external shocks without severe financial penalties.

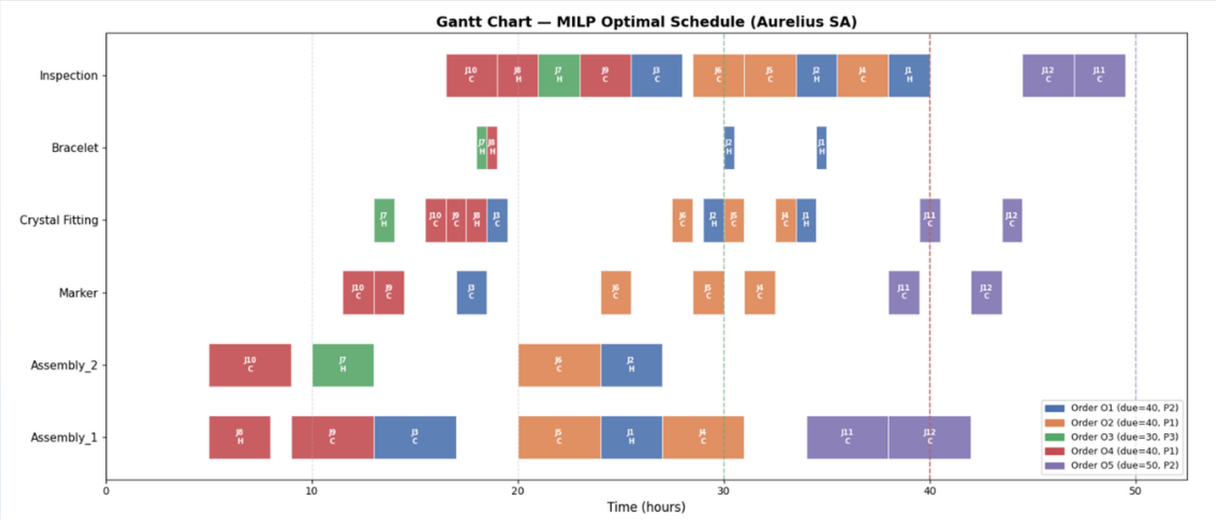
- Perfectly optimized capacity of 18,000 h/months
- 3235 Movement bottleneck
- Lead Time Fragility

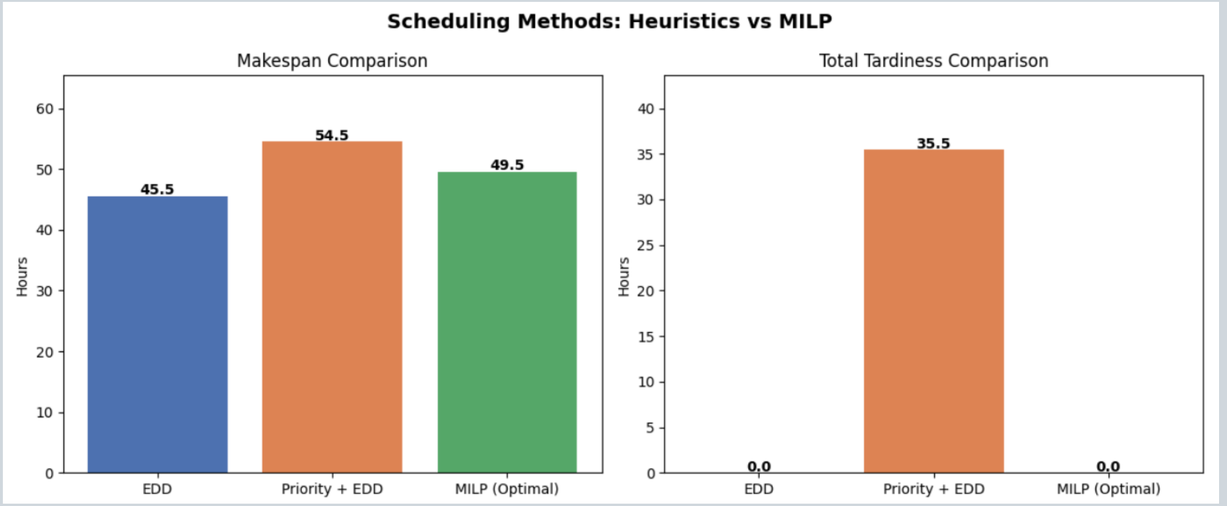
To scale profitably, Aurelius SA must transition from a strategy of pure "cost minimization" to aggressive "constraint management". By unblocking the 3235 Movement and fiercely protecting supplier lead times, the company can develop true operational agility.

PART 3

Base scheduling

- 12 individual watch jobs across 5 stations
- Two heuristics tested: EDD and Priority+EDD
- Full MILP optimization model





- EDD achieves zero tardiness in 45.5 hrs. MILP confirms optimality , all 5 orders on time

Disruptions :

1. Zero-wait Policy
2. Crystal fitting breakdown
3. Cleaning time
4. VIP order



PART 3

Comparison of disruptions scenarios

Model	Type	Cmax (hrs)	Total Tardiness (hrs)	Orders On Time
EDD Hybrid Zero-Wait	Heuristic	45.500000	0.0	5/5
MILP Zero-Wait	Optimal	47.000000	0.0	5/5
EDD Cleaning 10mn	Heuristic	45.666667	0.0	5/5
MILP Cleaning 10mn	Optimal	47.170000	0.0	5/5
EDD Cleaning 20mn	Heuristic	46.166663	0.5	4/5
MILP Cleaning 20mn	Optimal	45.830000	0.0	5/5
EDD Breakdown	Heuristic	54.500000	14.0	3/5
MILP Breakdown	Optimal	54.500000	14.0	3/5
EDD + VIP O6	Heuristic	47.000000	2.0	4/6
MILP + VIP O6	Optimal	48.000000	2.0	5/6

- EDD performs efficiently in stable production conditions, achieving low makespan and zero tardiness in standard scenarios.
- MILP demonstrates stronger robustness under most disruption scenarios, achieving zero tardiness under cleaning constraints and better VIP order protection, though both approaches are equally impacted by the Crystal Fitting breakdown.
- MILP becomes the preferable approach when delivery reliability is critical, despite requiring higher computational effort than EDD.

→ Recommendation: Aurelius SA should use EDD for routine stable production and switch to MILP when facing disruptions, cleaning constraints, or VIP order insertions where delivery reliability is critical.

Appendix

01 Level Strategy

02 Chase Strategy

03 MILP

04 EOQ + sensitivity analysis

05 Multi LS-U

06 Multi LS-C

07 MRP + Sensitivity Analysis

08 EDO

07 MILP

08 Disruptions

Problem Framing

The company must satisfy daily demand while deciding:

- *workforce levels*
- *hiring and firing decisions*
- *regular production*
- *overtime usage in eligible months*
- *subcontracting in eligible months*
- *inventory levels*

The objective is to minimize the cost over the planning horizon, it includes:

- *regular workforce wages*
- *hiring costs*
- *layoff costs*
- *inventory holding costs*
- *overtime costs*
- *subcontracting costs*

Constraints to respect:

- **Workforce balance**
- **Overtime and subcontracting only allowed up to a certain level and for specific months**

Decision Variables

- W_t: Number of workers employed
- H_t: Number of workers hired
- F_t: Number of workers laid off
- P_t: Regular production quantity
- I_t: Inventory level at the end of the month
- OT_t: Overtime production quantity
- S_t: Subcontracted quantity

Objective Function

Minimize total costs over the 12-month horizon:

$$\min \sum_{t=1}^{12} (7000W_t + 50000H_t + 25000F_t + 1000I_t + 14000OT_t + 15000S_t)$$

Workforce Balance Constraints

For the initial month ($t = 1$):

$$W_1 = 90 + H_1 - F_1$$

For subsequent months ($t = 2, \ldots, 12$):

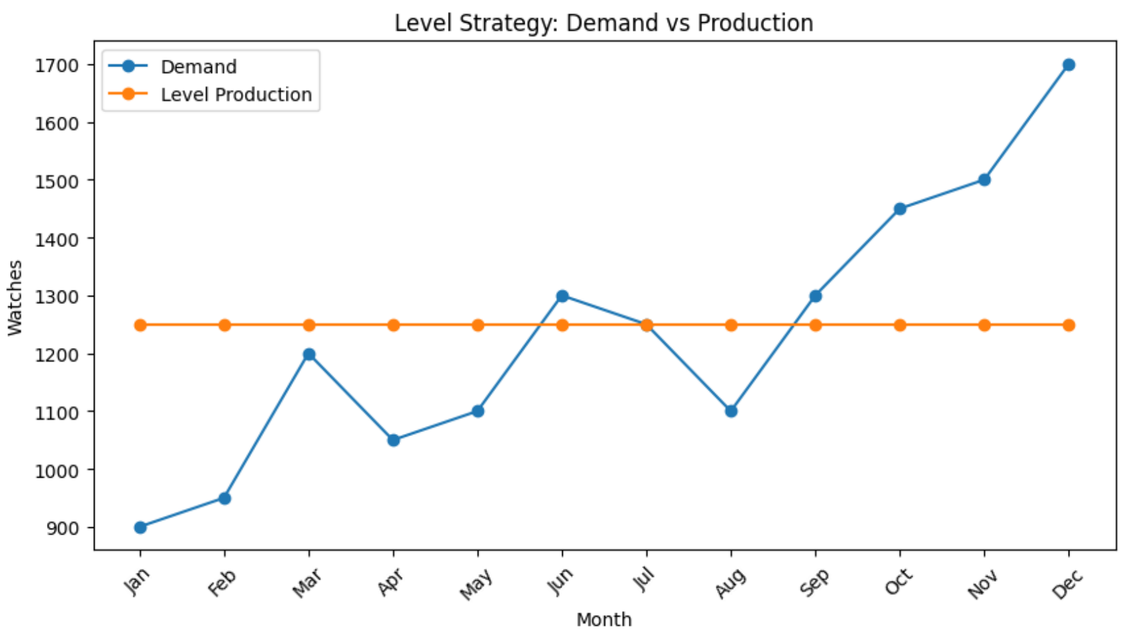
$$W_t = W_{t-1} + H_t - F_t$$

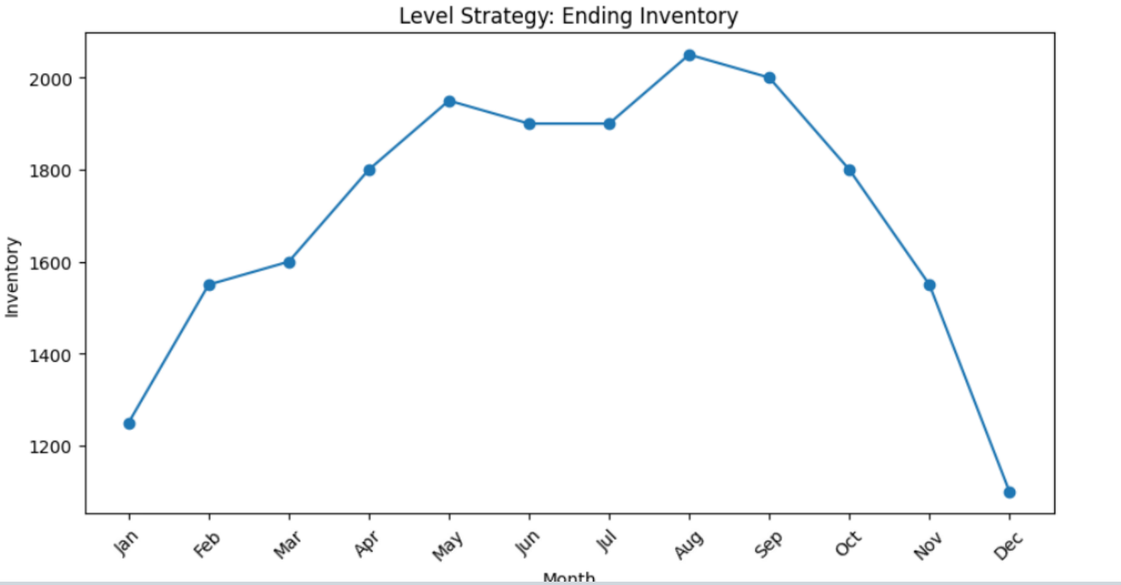
Appendix - Part 1 - Aggregate Planning

Level Strategy

Month	Demand	Workforce	Hiring	Firing	Production	Ending Inventory
Jan	900	125	35	0	1250	1250
Feb	950	125	0	0	1250	1550
Mar	1200	125	0	0	1250	1600
Apr	1050	125	0	0	1250	1800
May	1100	125	0	0	1250	1950
Jun	1300	125	0	0	1250	1900
Jul	1250	125	0	0	1250	1900
Aug	1100	125	0	0	1250	2050
Sep	1300	125	0	0	1250	2000
Oct	1450	125	0	0	1250	1800
Nov	1500	125	0	0	1250	1550
Dec	1700	125	0	0	1250	1100

Total cost = 32,700,000





Trade-off ⟶ workforce stability and inventory cost :

Avoiding repeated hiring and firing can be desirable in a luxury manufacturing environment where skills are valuable and training is expensive. But this strategy relies heavily on inventory to absorb variations, leading to substantial holding costs.

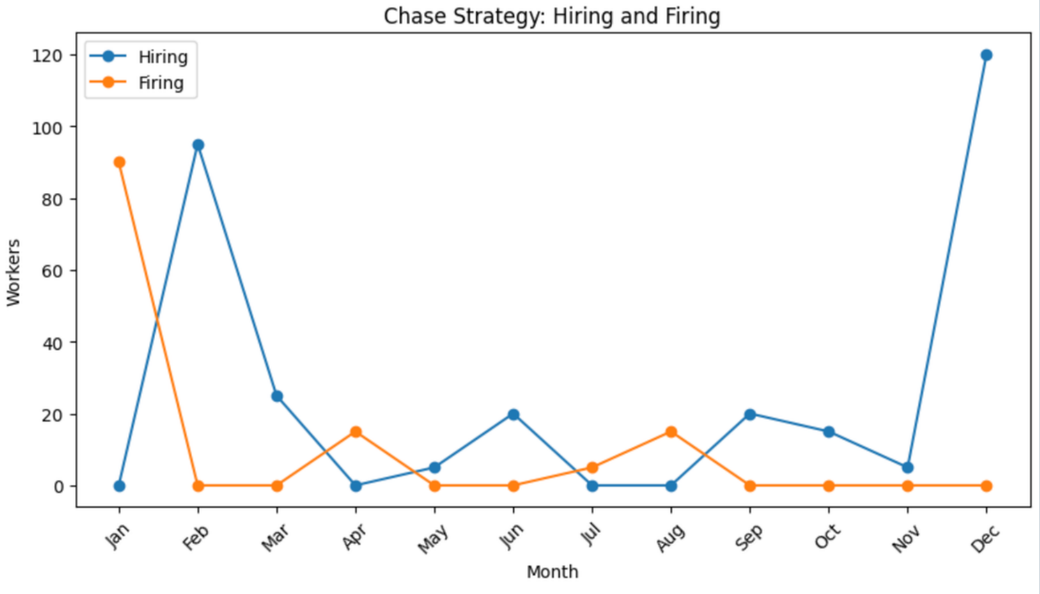
Appendix - Part 1 - Aggregate Planning

Chase Strategy

=== CHASE STRATEGY ===

Month	Demand	Workforce	Hiring	Firing	Production	Ending Inventory	Total Cost
Jan	900	0	0	90	0	0	2250000
Feb	950	95	95	0	950	0	5415000
Mar	1200	120	25	0	1200	0	2090000
Apr	1050	105	0	15	1050	0	1110000
May	1100	110	5	0	1100	0	1020000
Jun	1300	130	20	0	1300	0	1910000
Jul	1250	125	0	5	1250	0	1000000
Aug	1100	110	0	15	1100	0	1145000
Sep	1300	130	20	0	1300	0	1910000
Oct	1450	145	15	0	1450	0	1765000
Nov	1500	150	5	0	1500	0	1300000
Dec	1700	270	120	0	2700	1000	8890000






```
--- Annual Cost Breakdown ---
  Wage Cost    :   10,430,000 CHF
  Hire Cost    :   15,250,000 CHF
  Fire Cost    :    3,125,000 CHF
  Holding Cost :    1,000,000 CHF
```

Total cost = 29,805,000 CHF

Trade-off between workforce stability and inventory cost :

The inventory remains low which limits holding costs. However, the company must frequently hire and lay off workers, which is particularly costly in a luxury manufacturing environment where training is expensive and craftsmanship is valuable.

Appendix - Part 1 - Aggregate Planning

MILP

We solved four versions of the aggregate production planning model with increasing levels of flexibility.
The goal is to understand how different operational levers (overtime and subcontracting) impact total cost and production decisions.

Model	Total Cost
No overtime or subcontracting	20,280,000
Overtime only	19,607,000
Subcontracting only	20,280,000
Overtime and subcontracting	19,607,000

→ Overtime (1,400 CHF/unit) is cheaper than hiring (≈5,700 CHF/unit), so the model uses it in Sep and Dec. Subcontracting (15,000 CHF/unit) remains too expensive and is never used.

--- Monthly Plan ---

Month	Demand	Workforce	Hired	Fired	Production	Overtime	Subcontracted	Ending Inventory
Jan	900	80	0	10	800.0	0.0	0.0	800
Feb	950	80	0	0	800.0	0.0	0.0	650
Mar	1200	80	0	0	800.0	0.0	0.0	250
Apr	1050	80	0	0	800.0	0.0	0.0	0
May	1100	115	35	0	1150.0	0.0	0.0	50
Jun	1300	125	10	0	1250.0	0.0	0.0	0
Jul	1250	125	0	0	1250.0	0.0	0.0	0
Aug	1100	125	0	0	1250.0	0.0	0.0	150
Sep	1300	126	1	0	1260.0	4.0	0.0	114
Oct	1450	173	47	0	1730.0	0.0	0.0	394
Nov	1500	173	0	0	1730.0	0.0	0.0	624
Dec	1700	173	0	0	1730.0	346.0	0.0	1000

Sensitivity	Initial	New	New Total Cost
Overtime	**1400**	**700**	**19,362,000**
Subcontracting	**15,000**	**1000**	**18,712,400**
Overtime capacity	**20%**	**50%**	**18,733,000**
Subcontracting limit	**300**	**1000**	**19,362,000**

Limitation: Hiring and firing costs may not fully capture the loss of tacit craft knowledge in a luxury watchmaking context.

Appendix - Part 1 - Inventory Management

Economic Order Quantity

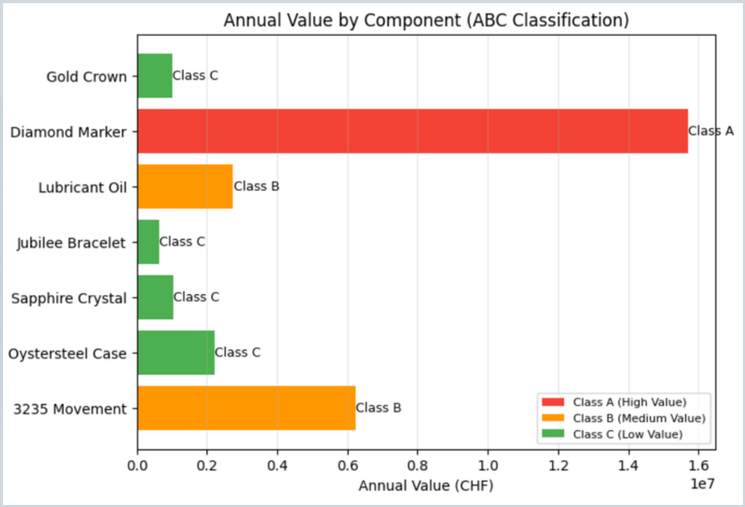
Service level = 95% — EOQ minimizes total ordering + holding costs per component

Component	CV	Type	EOQ	Safety Stock	ROP	Policy
3235 Movement	0.25	Moderate	81	45	901	EOQ + ROP + SS
Oystersteel Case	0.28	Moderate	127	47	730	EOQ + ROP + SS
Sapphire Crystal	0.25	Moderate	131	32	459	EOQ + ROP + SS
Jubilee Bracelet	0.12	Stable	88	10	290	EOQ + ROP
Lubricant Oil (g)	0.27	Moderate	428	253	2,364	EOQ + ROP + SS
Diamond Marker	0.11	Stable	139	38	1,042	EOQ + ROP
Gold Crown	0.38	Moderate	83	32	352	EOQ + ROP + SS

CV thresholds used to classify demand:

- **CV < 0.2 → Stable :** EOQ + ROP (no safety stock needed)
- **0.2 < CV < 0.5 → Moderate** : EOQ + ROP + Safety Stock
- **CV > 0.5 → Erratic** : none found in our data

Component	Orders/yr	Ordering cost	Holding cost	Total/yr (CHF)
3235 Movement	64.5	580,762	1,234,133	1,814,895
Oystersteel Case	43.6	305,077	531,820	836,897
Sapphire Crystal	39.4	157,790	235,359	393,149
Jubilee Bracelet	33.2	116,267	141,812	258,080
Lubricant Oil	128.5	128,469	280,534	409,002
Diamond Marker	125.3	751,896	1,166,578	1,918,474
Gold Crown	35.0	175,160	307,670	482,830



Key observations:

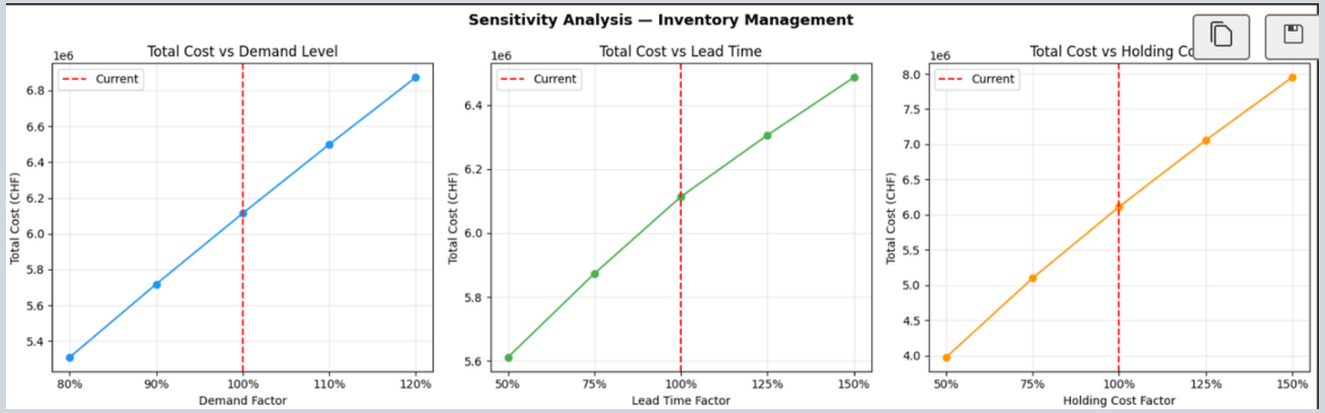
- **Diamond Marker has the highest annual cost (1,918,474 CHF) driven by very high ordering frequency (125×/yr) and high unit cost**
- **3235 Movement is costly due to 60-day lead time requiring large safety stock (45 units) → ROP = 901**
- **Lubricant Oil needs the largest safety stock (253g) due to high demand variability**
- **Jubilee Bracelet is the most stable component (CV=0.12) → minimal safety stock (10 units)**

Appendix - Part 1 - Inventory Management

Sensitivity Analysis

Sensitivity of EOQ to demand variation

Demand factor	3235 Mvt	Oyster	Sapphire	Jubilee	Lubric.	Diamond	Gold
80%	72	113	117	78	383	124	74
90%	76	120	124	83	406	132	79
100% ←	81	127	131	88	428	139	83
110%	84	133	137	92	449	146	87
120%	88	139	144	96	469	152	91



What the demand sensitivity shows:

- EOQ reacts proportionally to demand — a 10% demand increase raises EOQ by ~5% (square root relationship)
- Lubricant Oil is most sensitive in absolute terms (383→469 across the range) due to its very high daily demand
- Accurate demand forecasting directly reduces unnecessary inventory investment

What the sensitivity shows:

- ***Demand*** **↑ → cost linear, +20% demand = +13% total cost**
- ***Lead time*** **↑ → high impact, halving it saves ~500K CHF → prioritize 3235 Movement (60 days)**
- ***Holding cost*** **↑ → most sensitive, doubling it nearly doubles total cost — keep Class A stocks lean**

Problem Framing

The company must satisfy monthly demand while deciding:

- *when to set up the production line for each product model*
- *the production quantity for each model in each month*
- *the inventory levels to carry over to the next month*

The objective is to minimize the cost over the planning horizon, it includes:

- **variable production costs**
- **fixed setup costs**
- **inventory holding costs**

Constraints to respect:

- **Inventory balance**
- **Setup Logic : Production can only happen if the corresponding setup variable is activated (Yi,t=1)**

Additional constraint for LS-C

- **Capacity Limit : the total time required for both setup and regular production across all models cannot exceed the available 18000 hours in any given month t.**

Decision Variables

- $X_{i,t}$: Production quantity of product *i* for month *t*.
- $S_{i,t}$: Inventory level of product *i* at the end of month *t*.
- $Y_{i,t}$: Binary setup decision variable, equal to 1 if production of product *i* occurs in month *t* ($X_{i,t} > 0$), and 0, and 0 otherwise.

Objective Function

The objective is to minimize total costs over the planning horizon:

$$\min \sum_{i=1}^{I} \sum_{t=1}^{T} (P_{i,} X_{i,t} + Q_i Y_{i,t} + H_i S_{i,t})$$

Cost details:
$P_{i,}$: Unit production cost (doubles during peak months: June, Oct., Dec.)
Q_i: Fixed setup cost per batch **For LSC**
H_i: Monthly inventory holding cost per unit

Constraints

Inventory Balance:
Monthly demand must be met without backorders:

$$S_{i,t-1} + X_{i,t} = d_{i,t} + S_{i,t} \quad \forall i, t$$

Setup Logic:
Production only occurs if the setup is activated:

$$X_{i,t} \leq M Y_{i,t} \quad \forall i, t$$

Capacity Limit (LS-C only):
The total time required for setup and production cannot exceed the available hours:

$$\sum_i \alpha_i X_{i,t} + \sum_i \beta_i Y_{i,t} \leq 18000 \quad \forall t$$

α_i: Unit production time
β_i: Setup time per batch

Appendix - Part 2 - Lot-Sizing Models

Multi LS-U

Total cost = 59,900,000

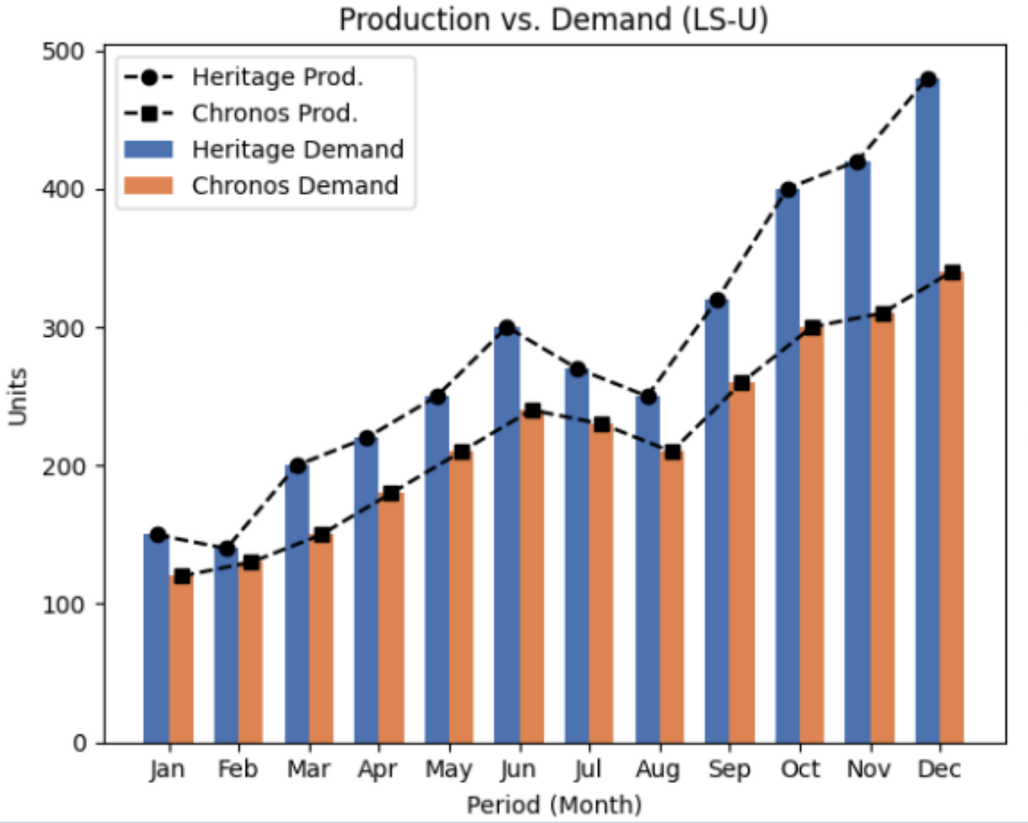
Produced in periods : 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 (for both model)

Chronos production details :

	Period	Demand	Production	Inventory	Setup
0	0	120	120.0	0.0	1.0
1	1	130	130.0	0.0	1.0
2	2	150	150.0	0.0	1.0
3	3	180	180.0	0.0	1.0
4	4	210	210.0	0.0	1.0
5	5	240	240.0	0.0	1.0
6	6	230	230.0	0.0	1.0
7	7	210	210.0	0.0	1.0
8	8	260	260.0	0.0	1.0
9	9	300	300.0	0.0	1.0
10	10	310	310.0	0.0	1.0
11	11	340	340.0	0.0	1.0

Heritage production details :

	Period	Demand	Production	Inventory	Setup
0	0	150	150.0	0.0	1.0
1	1	140	140.0	0.0	1.0
2	2	200	200.0	0.0	1.0
3	3	220	220.0	0.0	1.0
4	4	250	250.0	0.0	1.0
5	5	300	300.0	0.0	1.0
6	6	270	270.0	0.0	1.0
7	7	250	250.0	0.0	1.0
8	8	320	320.0	0.0	1.0
9	9	400	400.0	0.0	1.0
10	10	420	420.0	0.0	1.0
11	11	480	480.0	0.0	1.0



Strict **Lot-for-Lot** strategy: production exactly matches demand each month, resulting in zero inventory and monthly setups for both products
⟶ holding costs are highly punitive compared to setup costs

Appendix - Part 2- Lot-Sizing Models

Multi LS-C

Total cost = 60,118,000

zero-production for June, October and December, corresponding to the months having peak-time costs.

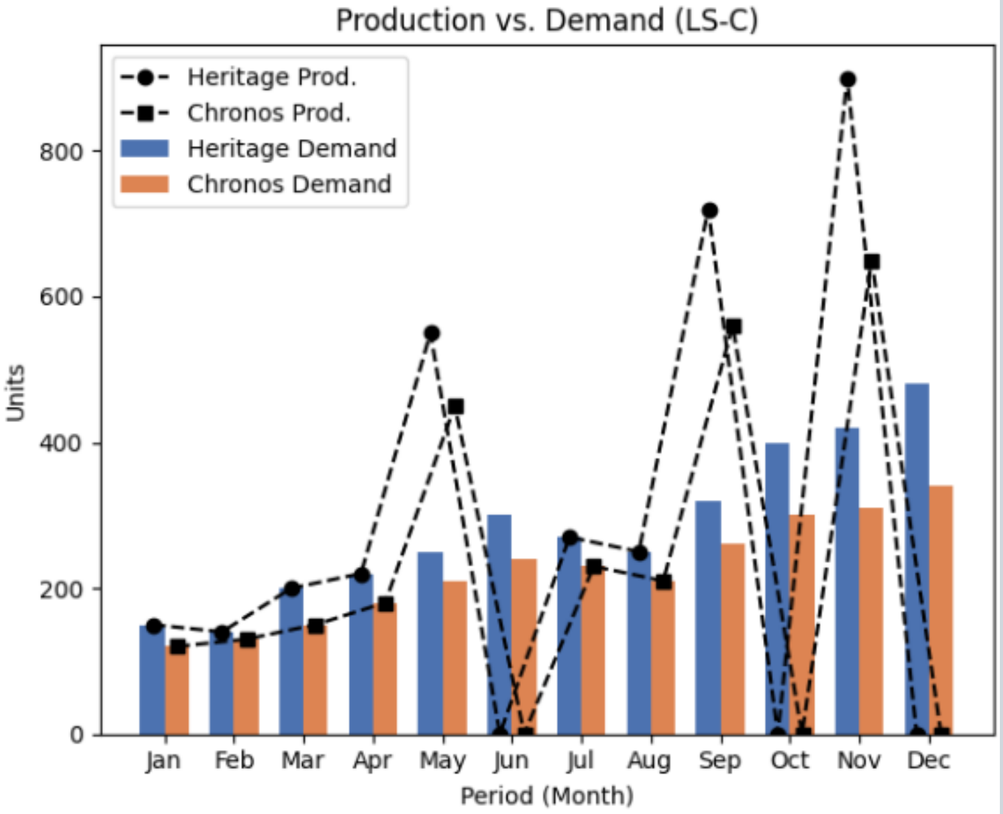
Produced in periods : 1, 2, 3, 4, 5, 7, 8, 9, 11 (for both model)

Chronos production details :

	Period	Demand	Production	Inventory	Setup
0	0	120	120.0	0.0	1.0
1	1	130	130.0	0.0	1.0
2	2	150	150.0	0.0	1.0
3	3	180	180.0	0.0	1.0
4	4	210	450.0	240.0	1.0
5	5	240	0.0	0.0	0.0
6	6	230	230.0	0.0	1.0
7	7	210	210.0	0.0	1.0
8	8	260	560.0	300.0	1.0
9	9	300	0.0	0.0	0.0
10	10	310	650.0	340.0	1.0
11	11	340	0.0	0.0	0.0

Heritage production details :

	Period	Demand	Production	Inventory	Setup
0	0	150	150.0	0.0	1.0
1	1	140	140.0	0.0	1.0
2	2	200	200.0	0.0	1.0
3	3	220	220.0	0.0	1.0
4	4	250	550.0	300.0	1.0
5	5	300	0.0	0.0	0.0
6	6	270	270.0	0.0	1.0
7	7	250	250.0	0.0	1.0
8	8	320	720.0	400.0	1.0
9	9	400	0.0	0.0	0.0
10	10	420	900.0	480.0	1.0
11	11	480	0.0	0.0	0.0



Lot-for-lot approach no longer viable ⟶ strategic shift to **anticipation inventory** :
Inventory pre-built to allow zero-production during the peak-months (holding cost cheaper than penalty of peak-months)

- Additional cost of 218,000 represents the necessary holding costs to smooth out production, respect the 18,000-hour capacity limit, and bypass the financial penalties of the peak season.

Problem Framing

The company must satisfy monthly demand while deciding:

- *Lot-Sizing and Timing: Calculating the optimal quantities and schedules for each of the 7 components*
- *Inventory Management: Tracking initial inventories and calculating projected on-hand stock across all periods to avoid shortages.*
- *Component Allocation: Determining the exact distribution of Oystersteel Cases between internal manufacturing needs and external demand.*

The objective is to minimize the cost over the planning horizon, it includes:

- **variable production costs**
- **fixed setup costs**
- **inventory holding costs**

Constraints to respect:

- **Inventory balance**
- **Setup Logic : Production can only happen if the corresponding setup variable is activated (Yi,t=1)**

Decision Variables

- $X_{i,t}$: Production quantity of product *i* for month *t*.
- $S_{i,t}$: Inventory level of product *i* at the end of month *t*.
- $Y_{i,t}$: Binary setup decision variable, equal to 1 if production of product *i* occurs in month *t* ($X_{i,t} > 0$), and 0, and 0 otherwise.

Objective Function

The objective is to minimize total costs over the planning horizon:

$$\min \sum_{i=1}^{I} \sum_{t=1}^{T} (P_{i,t} X_{i,t} + Q_i Y_{i,t} + H_i S_{i,t})$$

Cost details:
$P_{i,t}$: Unit production cost (doubles during peak months: June, Oct., Dec.)
Q_i: Fixed setup cost per batch **For LSC**
H_i: Monthly inventory holding cost per unit

Production & Inventory Constraints

Inventory Balance Equations
For all items *i* and all periods $t = 1, \ldots, 12$:

$$s_{it-1} + x_{it} = d_{it} + \sum_{j \in P(i)} r_{ij} x_{jt} + s_{it}$$

s_{it}: Inventory level of item *i* at the end of period *t*
x_{it}: Production quantity of item *i* in period *t*
d_{it}: Demand for item *i* in period *t*
r_{ij}: Bill of Materials usage rate (units of *i* per parent *j*)
x_{jt}: Production of parent item *j*
$P(i)$: Set of parent items for item *i*

Production Setup Logic
For all items *i* and all periods $t = 1, \ldots, 12$:

$$x_{it} \leq M \cdot y_{it}$$

M: A very large positive number (Big M)
y_{it}: Binary variable (1 if item *i* is produced in period *t*, 0 otherwise)

Initial Inventory Condition
For all items *i* at period $t = 0$: $s_{i0} = \text{Initial}_i$

Initial_i: Known initial stock data for item *i*.

Appendix - Part 2- MRP

MRP

For this analysis, we solved in total 4 MRP :

Total costs	No external demand	External demand	Difference
Sequential	129,575,010	129,589,570	14,560
Integrated	129,424,860	129,692,970	268,110
Number of setups	sequential : 5 integrated : 5	sequential : 6 integrated : 5	sequential : 1 integrated : 0
Difference	150,150	103,400	

Sequential to integrated :

- Global synergies uncovered, which lead to a cost reduction
- Better anticipation of the demand better ⟶ workload smoothed across the year
- Dynamic Batching Strategy : Forecasting allows to safely reduce both machine setup frequenc and inventory holding levels

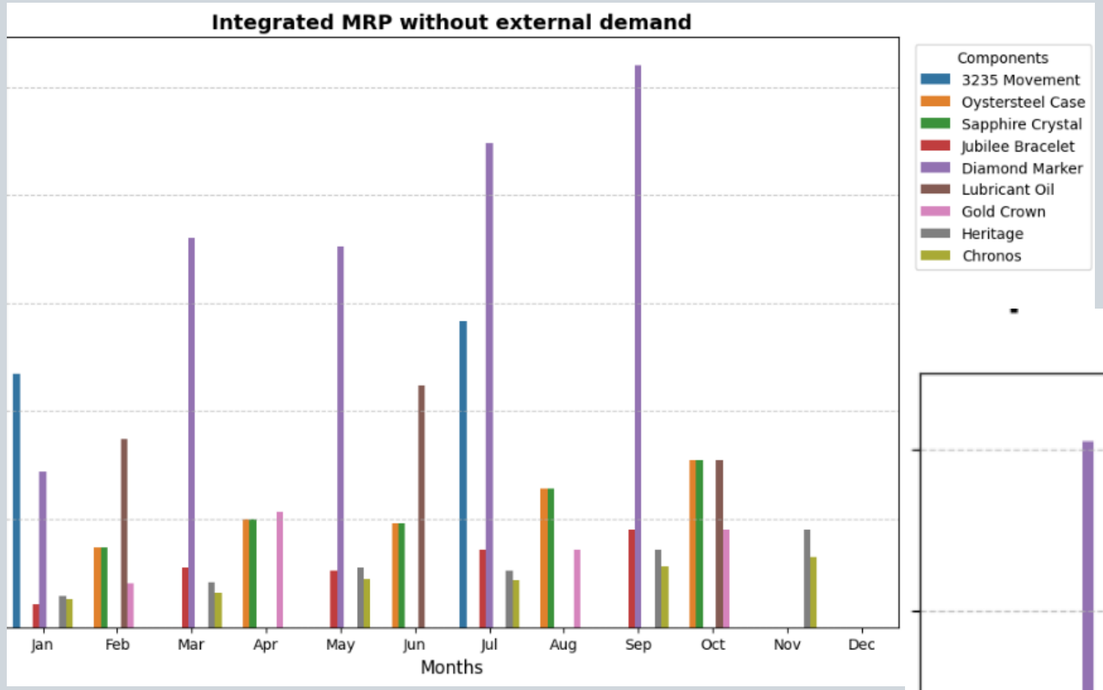
No external demand to External demand : low but disruptive volume of 118 additional units spread across January to June

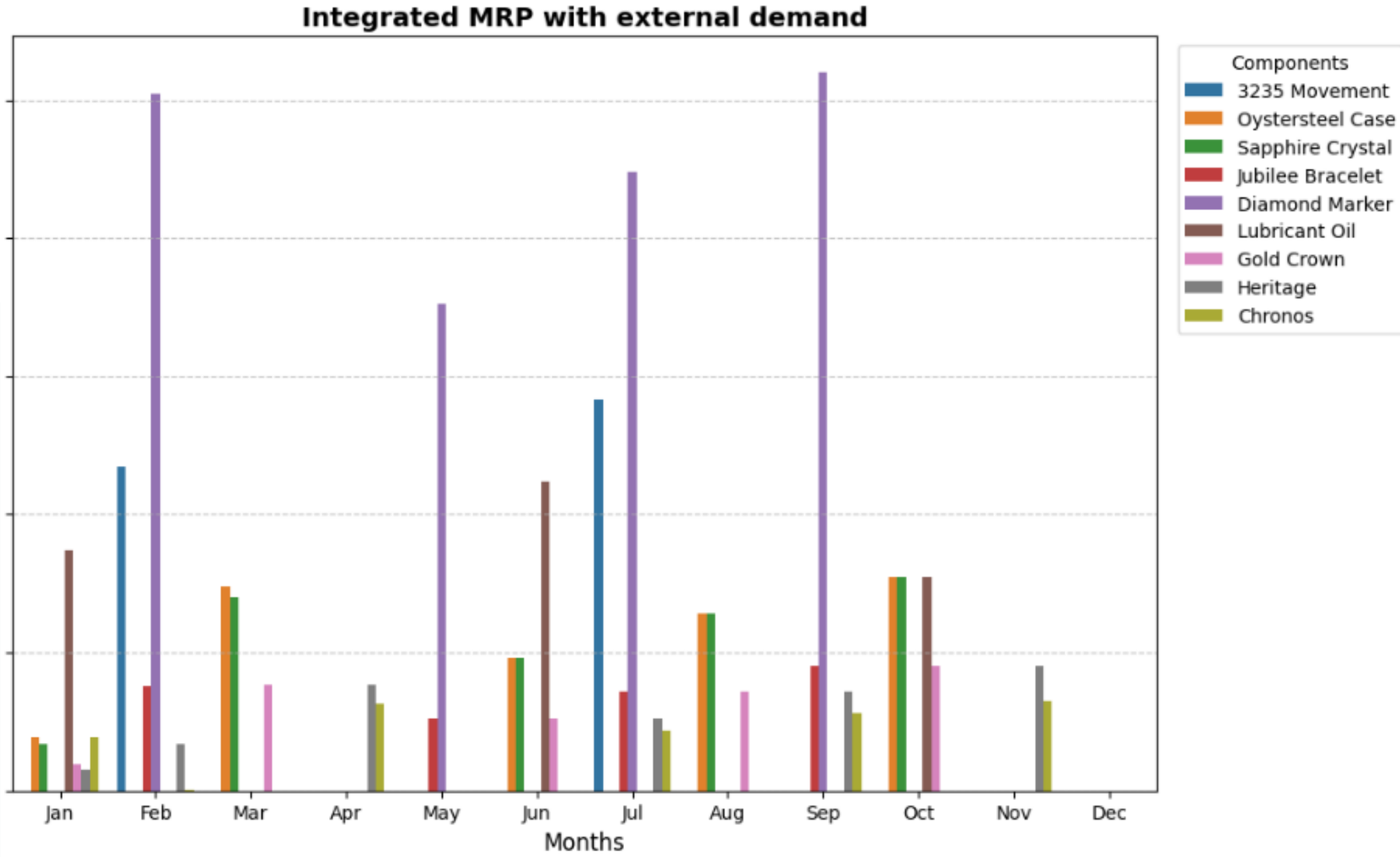
- Cost increase
- Production starts earlier (January instead of February)
- Higher inventory (slight overproduction in January and March, to avoid new triggering costly new setups)

External demand trade-offs:

- Fulfilling high external demand will saturate our production line ⟶ stockouts for internal use, ⟶ blocking the final assembly
- Internal assembly should be prioritized : The core business value and profit margins derived from selling finished luxury watches are probably higher than those from selling individual raw components
- Allocation of components to external clients only if :
  revenue generated from external sales ≥ marginal cost increase

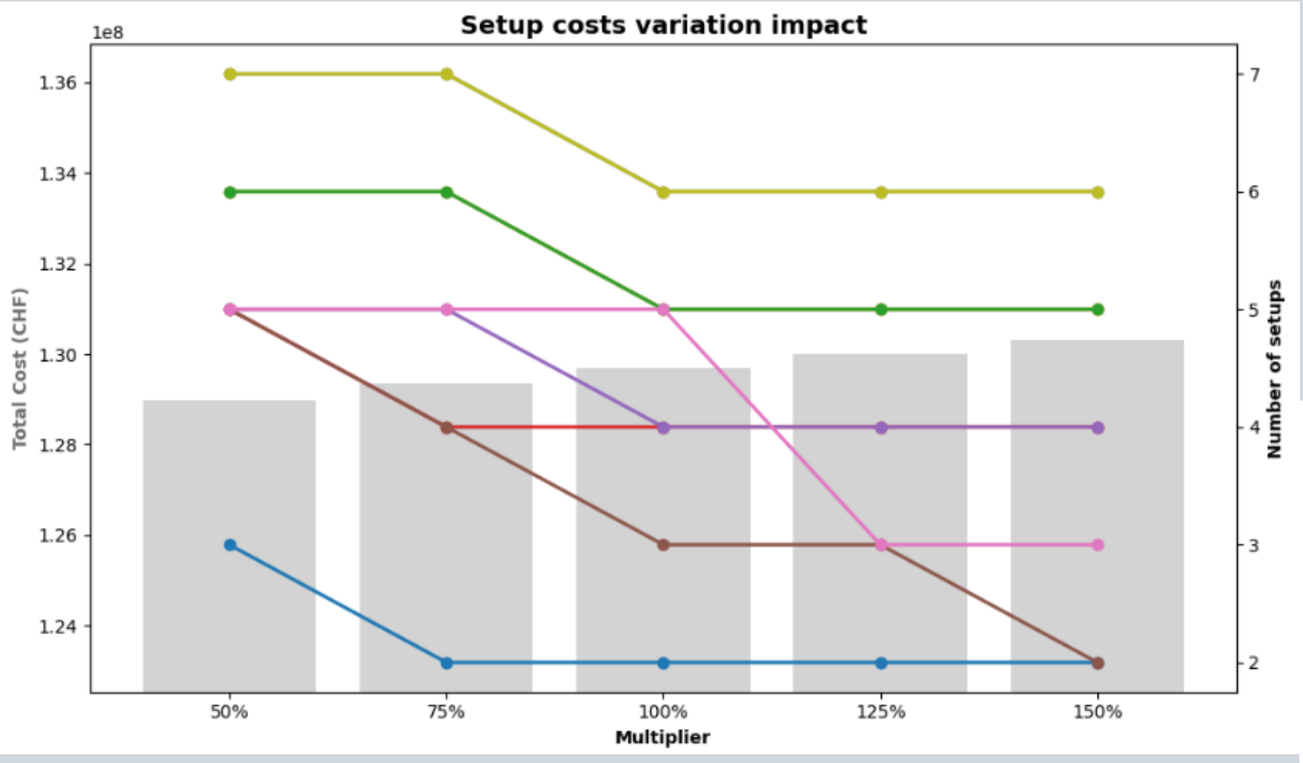
Optimal Production Plan



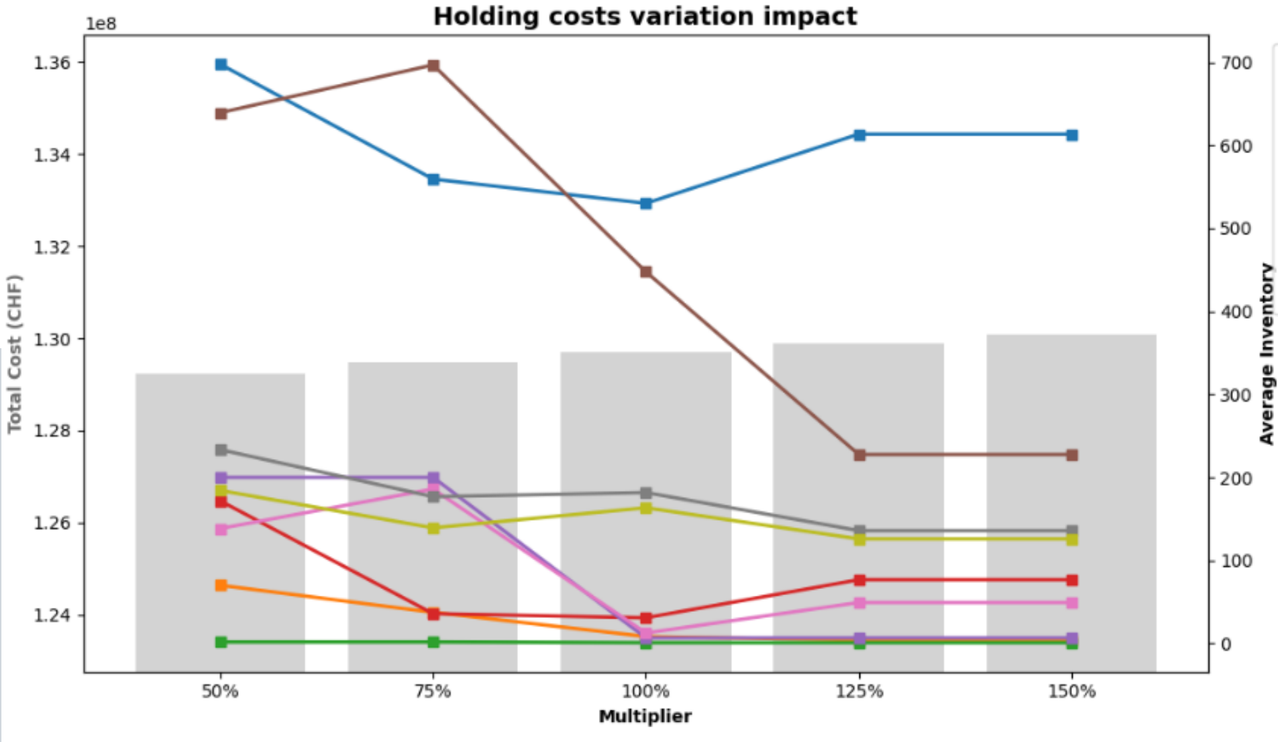


Appendix - Part 2- MRP

Sensitivity Analysis (1)



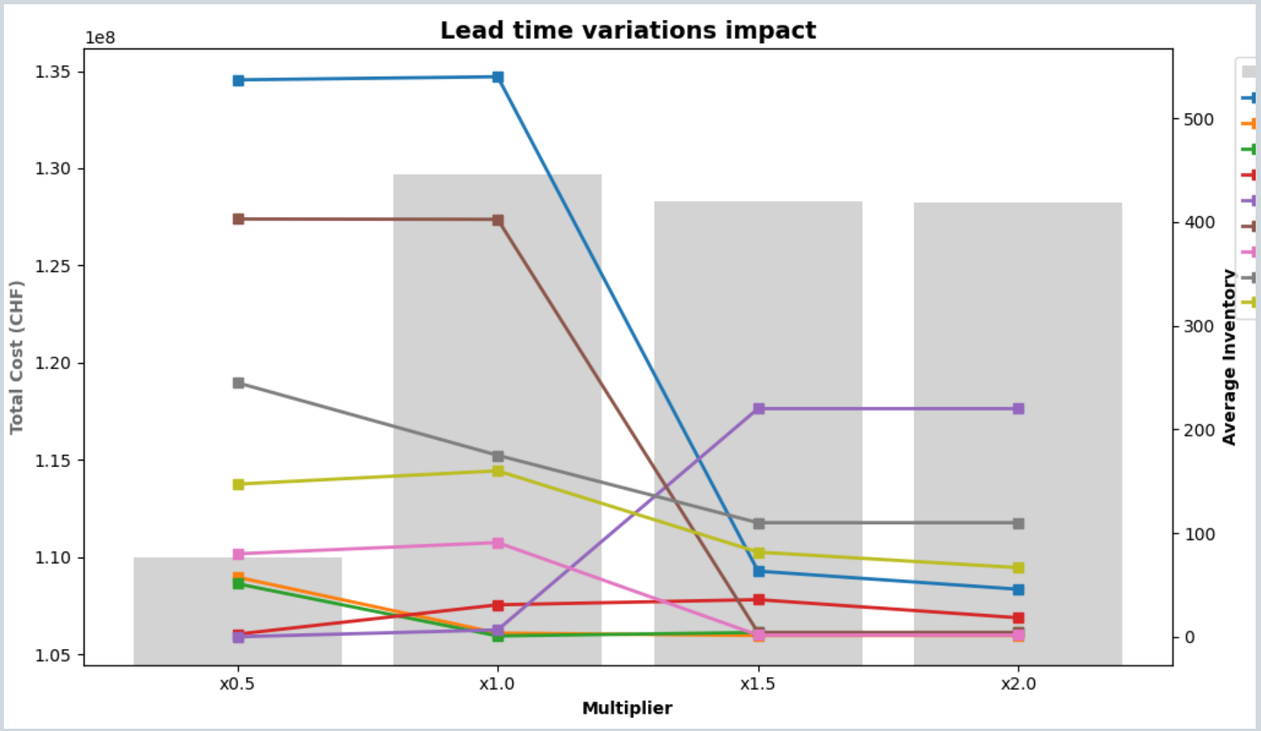


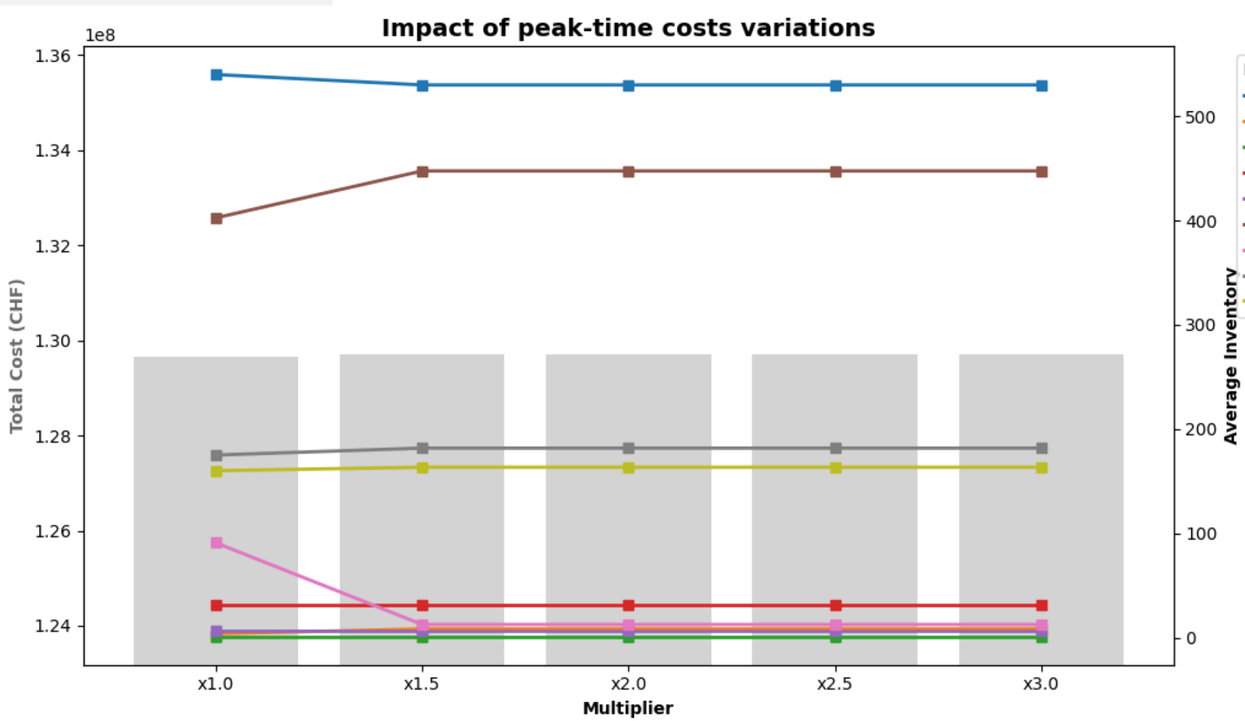


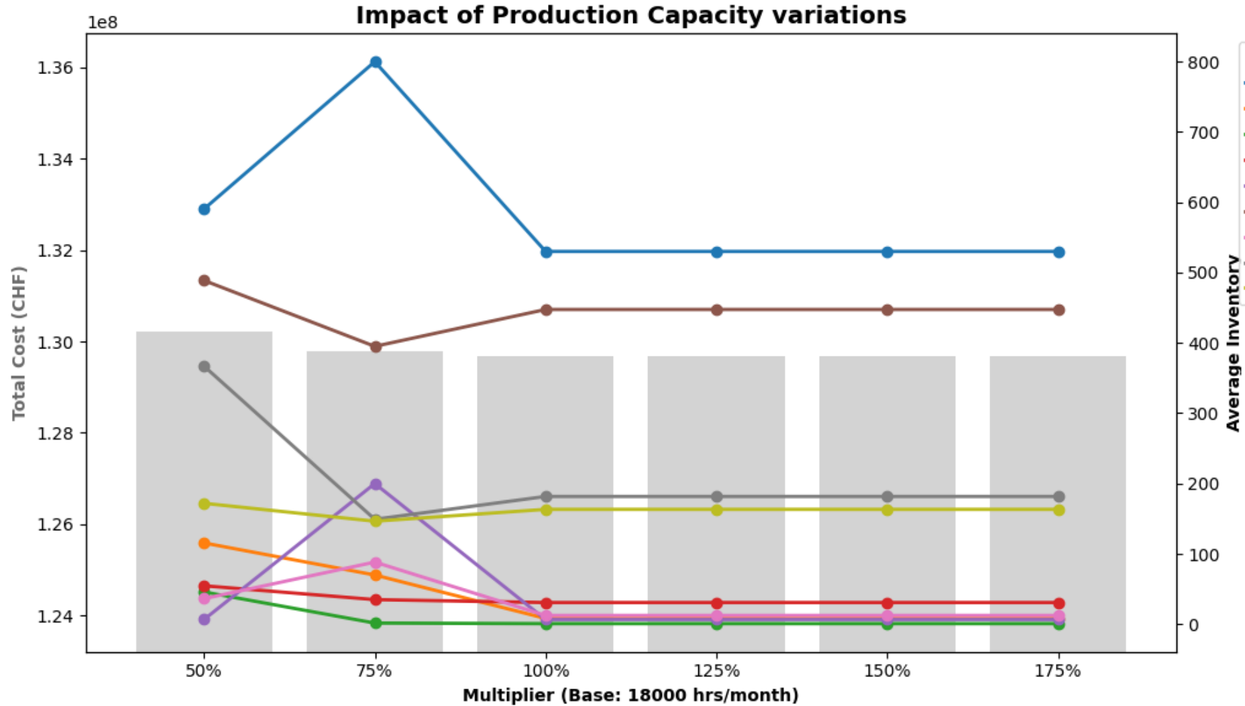
Appendix - Part 2- MRP

Sensitivity Analysis (2)









Appendix - Part 3- Scheduling

Problem Framing

Variables & Objective function

- Start(j, k): the time at which job j begins operation k
- x(j1, j2, s): a binary variable equal to 1 if job j1 is processed before job j2 on station s, and 0 otherwise

$$\min \sum w_o \cdot T_o$$

- $T_o = \max(0, C_o - d_o)$ is the tardiness of order o
- C_o is the completion time of order o (= the last job in that order to finish)
- d_o is the due date of order o
- w_o is the priority weight of order o (priority 1 orders are penalized more)

Constraints

- Release dates: no job can start before its order's release time.

$$Start(j,1) \geq r_j \quad \forall j$$

- Precedence (routing sequence): within a job, each operation must finish before the next one begins.

$$Start(j,k+1) \geq Start(j,k) + p(j,k) \quad \forall j,k$$

- No overlap on single-machine stations: two jobs cannot use the same station at the same time.

$$Start(j1,k) \geq Start(j2,k) + p(j2,k) \text{ or } Start(j2,k) \geq Start(j1,k) + p(j1,k)$$

- Assembly station (2 machines): up to 2 jobs can overlap at Assembly since there are 2 parallel machines.
- Non-negativity: all start times must be ≥ 0.

Appendix - Part 3- Scheduling

EDD Heuristic

Model & Results

```
# ============================================================
# HEURISTIC 1 — EDD (Earliest Due Date)
# Sort jobs purely by their due date.
# Jobs with the tightest deadline go first.
# ============================================================

jobs_edd = sorted(jobs, key=lambda j: (j["due"], j["release"]))

schedule_edd, completion_edd, tardiness_edd = run_schedule(jobs_edd)

# --- Results table ---
print("=== HEURISTIC 1: EDD ===\n")

results_edd = pd.DataFrame([{
    "Order":           oid,
    "Due":             orders[oid]["due"],
    "Priority":        orders[oid]["priority"],
    "Completion":      completion_edd[oid],
    "Tardiness (hrs)": tardiness_edd[oid],
    "On Time?":        "✅" if tardiness_edd[oid] == 0 else "❌"
} for oid in orders])

display(results_edd)

cmax_edd = max(completion_edd.values())
total_tardiness_edd = sum(tardiness_edd.values())
print(f"Makespan (Cmax):      {cmax_edd} hrs")
print(f"Total Tardiness:      {total_tardiness_edd} hrs")
print(f"Orders on time:       {sum(1 for t in tardiness_edd.values() if t == 0)}/5")
```

	Order	Due	Priority	Completion	Tardiness (hrs)	On Time?
0	O1	40	2	23.0	0	✅
1	O2	40	1	37.5	0	✅
2	O3	30	3	16.5	0	✅
3	O4	40	1	30.0	0	✅
4	O5	50	2	45.5	0	✅

EDD + Priority Heuristic

Model & Results

```
# ============================================================
# HEURISTIC 2 — Priority + EDD
# Sort jobs by priority first (1 = most urgent),
# then by due date to break ties within same priority.
# ============================================================

jobs_prio = sorted(jobs, key=lambda j: (j["priority"], j["due"]))

schedule_prio, completion_prio, tardiness_prio = run_schedule(jobs_prio)

# --- Results table ---
print("=== HEURISTIC 2: Priority + EDD ===\n")

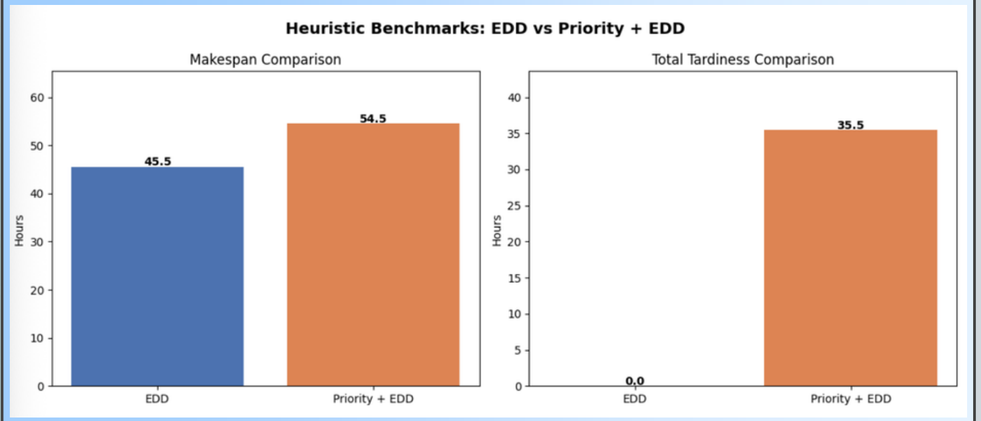
results_prio = pd.DataFrame([{
    "Order":           oid,
    "Due":             orders[oid]["due"],
    "Priority":        orders[oid]["priority"],
    "Completion":      completion_prio[oid],
    "Tardiness (hrs)": tardiness_prio[oid],
    "On Time?":        "✅" if tardiness_prio[oid] == 0 else "❌"
} for oid in orders])

display(results_prio)

cmax_prio = max(completion_prio.values())
total_tardiness_prio = sum(tardiness_prio.values())
print(f"Makespan (Cmax):      {cmax_prio} hrs")
print(f"Total Tardiness:      {total_tardiness_prio} hrs")
print(f"Orders on time:       {sum(1 for t in tardiness_prio.values() if t == 0)}/5")
```

Order	Due	Priority	Completion	Tardiness (hrs)	On Time?
O1	40	2	47.5	7.5	❌
O2	40	1	34.0	0.0	✅
O3	30	3	54.5	24.5	❌
O4	40	1	41.0	1.0	❌
O5	50	2	52.5	2.5	❌

Comparison of the 2 Heuristics



- *Makespan 45.5 VS 54.5*
- *tardiness 0 VS 35.5*
- *orders on time 5 VS 1*

Appendix - Part 3- Scheduling

Exact Method ; MILP

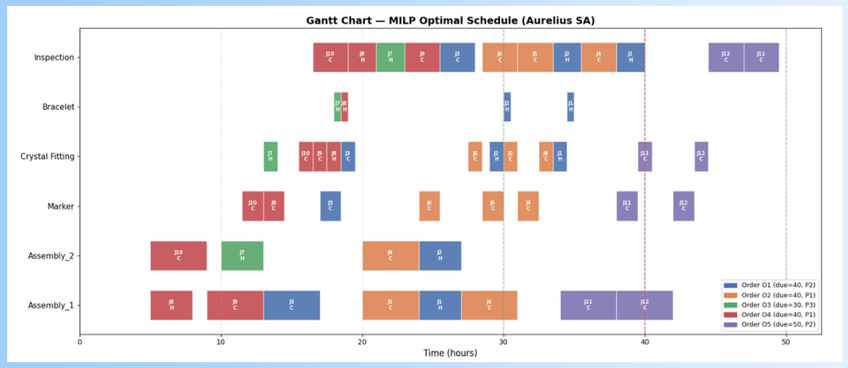
01 Model

```
b = seq[(id1, id2)][s1]
# Either j1 before j2:
model += start[id1][k1] + dur1 <= start[id2][k2] + M * (1 - b)
# Or j2 before j1:
model += start[id2][k2] + dur2 <= start[id1][k1] + M * b
```

```
# ============================================================
# REASSIGN ASSEMBLY JOBS TO MACHINES BASED ON TIMING
# The MILP may put all jobs on Assembly_1 — we redistribute
# them here for a clean visual display
# ============================================================
assembly_tasks = []
for j in jobs:
    jid = j["job_id"]
    for k, (station, duration) in enumerate(j["routing"]):
        if station == "Assembly":
            s = value(start_vars[jid][k])
            assembly_tasks.append({
                "job_id": jid,
                "start": round(s, 2),
                "end": round(s + duration, 2),
            })
```

02 Results and gant chart

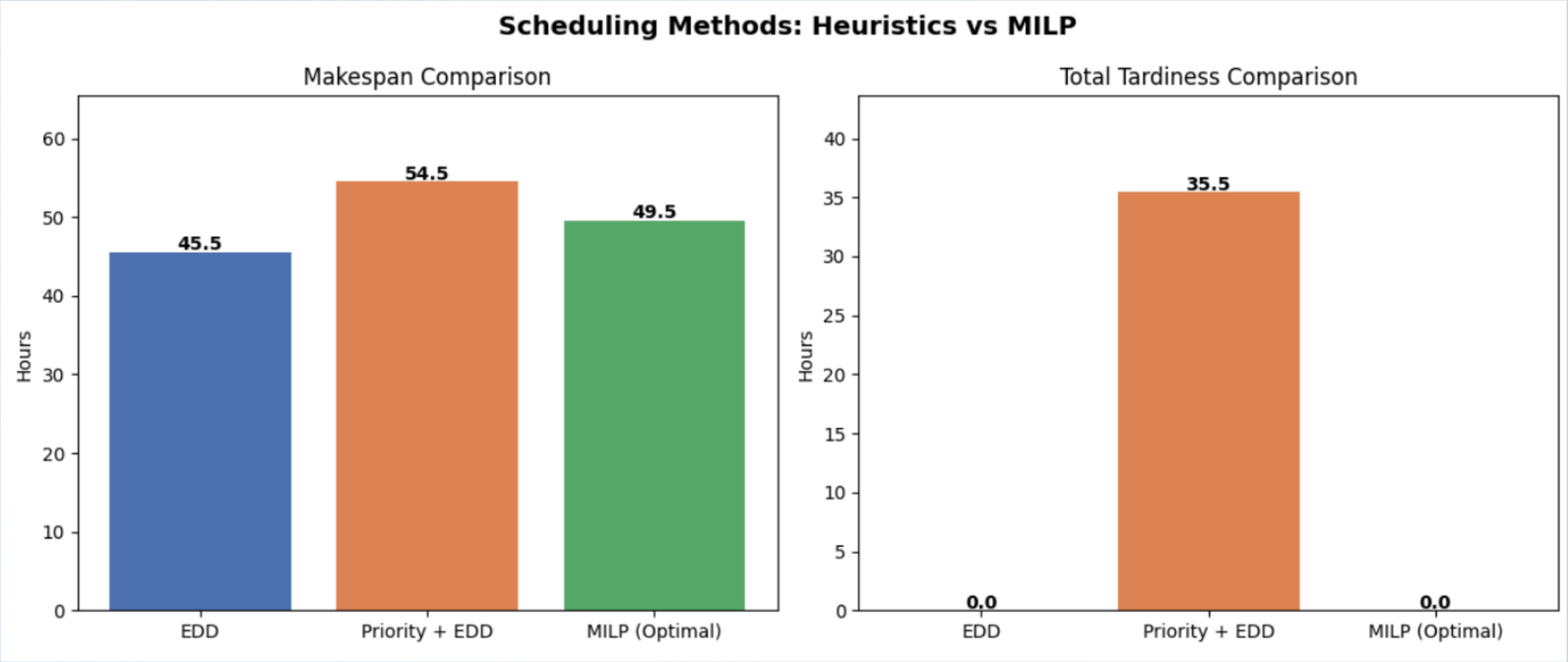
	Order	Due	Priority	Completion	Tardiness (hrs)	On Time?
0	O1	40	2	40.0	0.0	✅
1	O2	40	1	38.0	0.0	✅
2	O3	30	3	23.0	0.0	✅
3	O4	40	1	25.5	0.0	✅
4	O5	50	2	49.5	0.0	✅



EDD VS EDD + priority VS MILP

- EDD accidentally finds the shortest schedule (45.5 hrs) because its simple deadline-first rule happens to work perfectly on this dataset

- MILP guarantees zero tardiness with weighted priority protection, but isn't asked to minimize makespan, so it doesn't



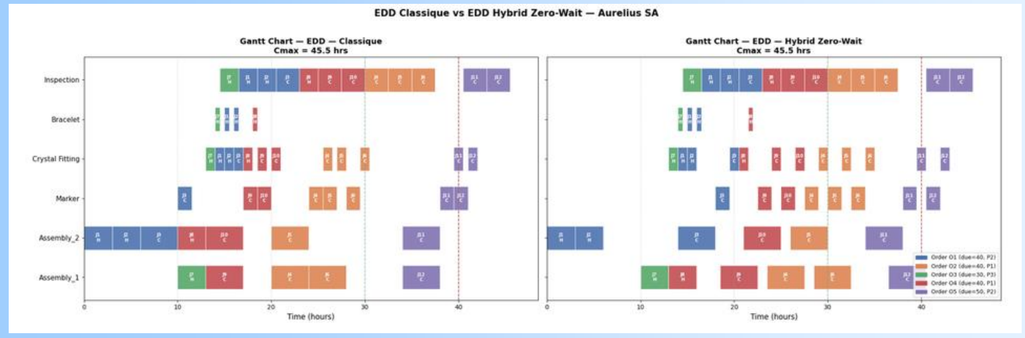
Disruptions :

1. Zero-wait Policy
2. Crystal fitting breakdown
3. Cleaning time
4. VIP order



1 Zero-wait Policy

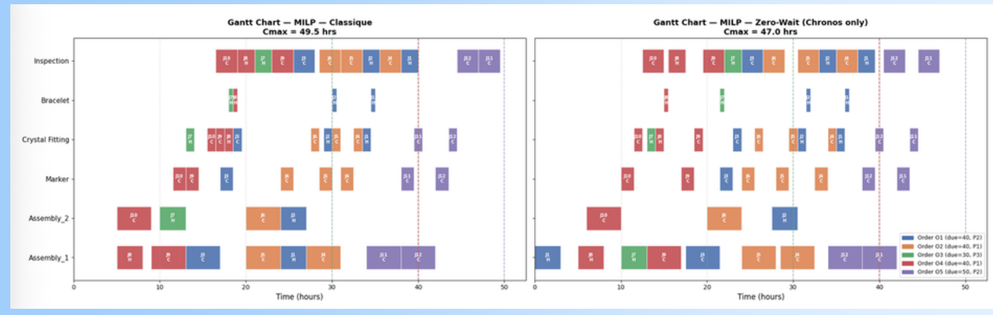
EDD



- Chronos watches require strict zero-wait flow, no delay allowed between consecutive operations
- Initial hybrid scheduler had waiting gaps fixed by separating feasibility checks from machine reservation
- Zero-wait increases sequencing rigidity — some machines
- stay intentionally idle to preserve uninterrupted flow
- - Key takeaway: zero-wait strongly impacts local sequencing

while maintaining the same global makespan as classic EDD

MILP



- Classical MILP allows waiting times —> mproves machine utilization & reduces conflicts
- Hybrid Zero-Wait MILP imposes strict continuity for Chronos watches —> no waiting between operations
- *Chronos* jobs become highly synchronized across Assembly, Crystal Fitting, Markers & Inspection
- Makespan improves despite added constraints —> 49.5 h to 47.0 h
- System keeps enough flexibility/capacity to absorb Zero-Wait constraints without major congestion
- Zero-Wait reduces local scheduling flexibility —> larger idle periods on some machines
- Chronos watches remain flexible —> act as adaptive buffers supporting overall performance
- Key takeaway —> premium Heritage continuity achieved while maintaining strong global efficiency

Appendix - Part 3- Scheduling

2 Cleaning time

EDD

10mn cleaning time

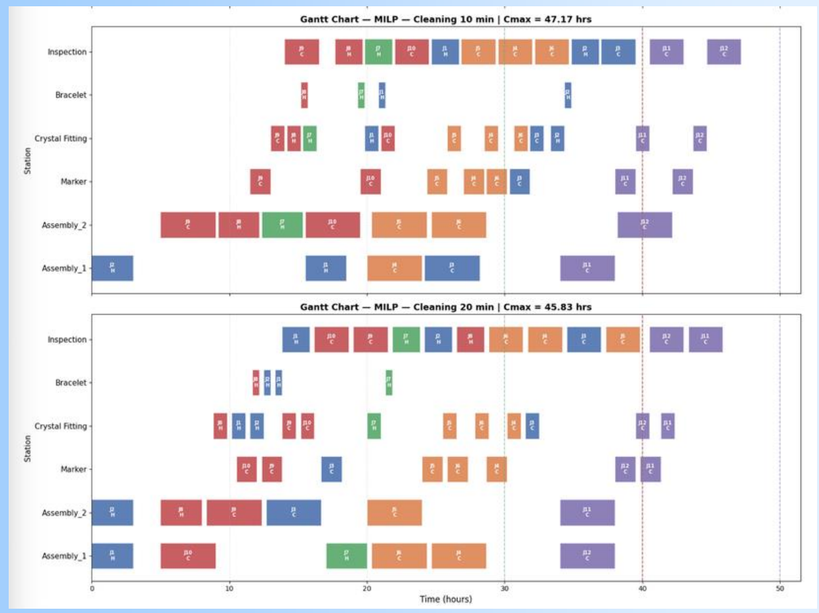
Order	Due	Priority	Completion	Tardiness (hrs)	On Time?
O1	40	2	23.500001	0	✅
O2	40	1	39.000003	0	✅
O3	30	3	16.500000	0	✅
O4	40	1	31.000002	0	✅
O5	50	2	45.666667	0	✅

20mn cleaning time

Order	Due	Priority	Completion	Tardiness (hrs)	On Time?
O1	40	2	23.999999	0.0	✅
O2	40	1	40.499997	0.5	❌
O3	30	3	16.500000	0.0	✅
O4	40	1	31.999998	0.0	✅
O5	50	2	46.166663	0.0	✅

• Cleaning times added between operations when watch type changes at a station
• 10-min cleaning → all 5 orders on time, makespan = 45.67 h
• 20-min cleaning → Order O2 (Priority 1) slips by 0.5 h, only 4/5 on time
• EDD is purely deadline-driven → cannot anticipate cumulative setup costs
• Cleaning penalties accumulate across stations → cascading delays on tight schedules
•Simple dispatching rule loses grip when setup constraints intensify

MILP



• 10-min cleaning → makespan improves from 49.0 h to 47.17 h vs classical MILP
• 20-min cleaning → makespan improves further to 45.83 h — counterintuitive result
- Heavier cleaning forces optimizer to batch similar watch families together
• Fewer transitions → less congestion, better station synchronization
• Cleaning constraints act as regulators → more balanced workload distribution
• Zero tardiness maintained in both cleaning scenarios → 5/5 orders on time

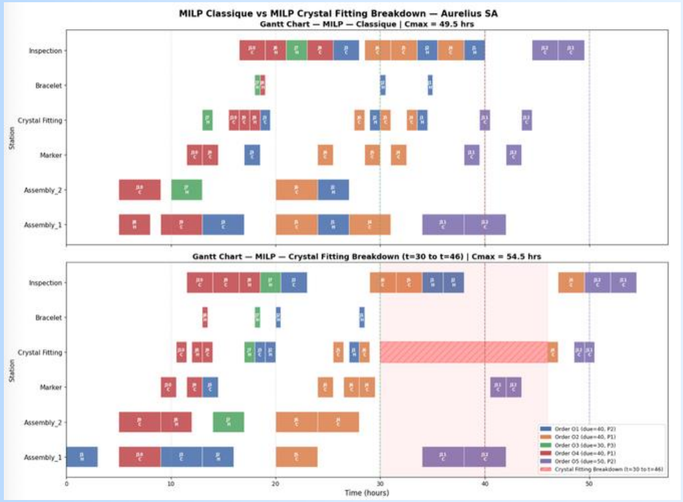
3 Crystal fitting breakdown

EDD

Order	Due	Priority	Completion	Tardiness (hrs)	On Time?
O1	40	2	23.0	0.0	✅
O2	40	1	49.5	9.5	❌
O3	30	3	16.5	0.0	✅
O4	40	1	30.0	0.0	✅
O5	50	2	54.5	4.5	❌

- *EDD baseline schedule: smooth and synchronized production flow with a makespan of 45.5 h.*
- *Disruption introduced: breakdown of the Crystal Fitting machine between t = 30 and t = 46.*
- *Main effect: operations scheduled during the failure window are postponed, creating cascading delays across downstream tasks.*
- *Most impacted jobs: Orders 2 and 4, due to their dependency on the Crystal Fitting workstation.*
- *Global impact: makespan increases from 45.5 h to 54.5 h (~20% increase), showing how a critical machine failure propagates delays through the*

MILP



- **Classical MILP under normal conditions —> balanced flow & makespan = 49.5 h**
- **Breakdown scenario —> Crystal Fitting unavailable from t = 30 to 46**
- **MILP dynamically reorganizes the schedule instead of simply delaying tasks**
- **Jobs resequenced —> some operations advanced, others strategically delayed**
- **Goal —> preserve feasibility & maintain resource synchronization**
- **Breakdown creates cascading delays, especially in Inspection & final assembly**
- **Despite a 16-hour breakdown, makespan rises only = 54.5 h**
- **Key takeaway —> predictive-reactive scheduling improves robustness through global rescheduling & workload balancing**

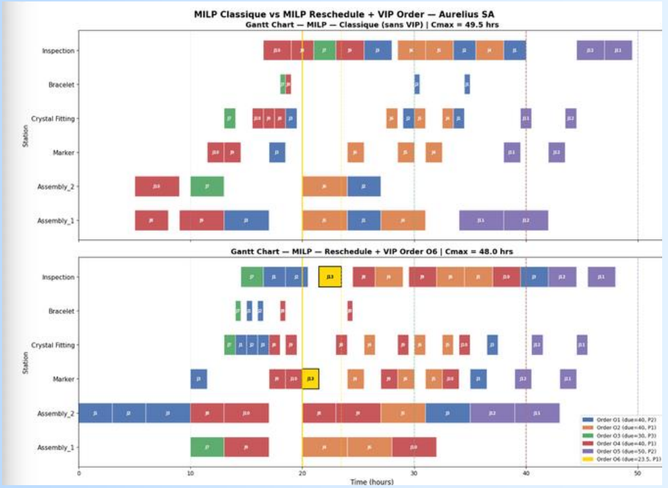
4 VIP order

EDD

Order	Due	Priority	Completion	Tardiness (hrs)	On Time?
O1	40.0	2	42.0	2.0	❌
O2	40.0	1	39.5	0.0	✅
O3	30.0	3	16.5	0.0	✅
O4	40.0	1	32.0	0.0	✅
O5	50.0	2	47.0	0.0	✅
O6	23.5	1	23.5	0.0	✅

- *Dynamic disruption introduced: a high-priority VIP order arrives at t = 20 during production execution.*
- *EDD reaction: the VIP job receives immediate priority, forcing a partial rescheduling of ongoing operations.*
- *Main impact: lower-priority jobs are postponed to free capacity on critical stations such as Marker and Inspection.*
- *System flexibility: despite the disruption, the production flow remains relatively stable and feasible.*
- *Global impact: makespan increases only slightly from 45.5 h to 47.0 h, showing the adaptability of EDD scheduling to urgent customer requests.*

MILP



- **Classical MILP under stable conditions —> makespan = 49.5 h**
- **Rescheduling scenario ⟶ urgent VIP order J13 arrives at t = 20**
- **MILP immediately prioritizes J13, especially on Marker & Inspection stations**
- **Schedule globally reorganized —> resequencing & workload redistribution**
- **Global reoptimization maintains synchronization & limits congestion**
- **Rescheduled plan slightly improves makespan —> = 48.0 h**
- **VIP order integrated without degrading system performance**
- **Key takeaway —> predictive-reactive scheduling ensures robustness & adaptability**

Thank You for your

